TUESDAY | Resistors + LED Circuit Lab

Grades 9-10 | Name: _____ Date: _____

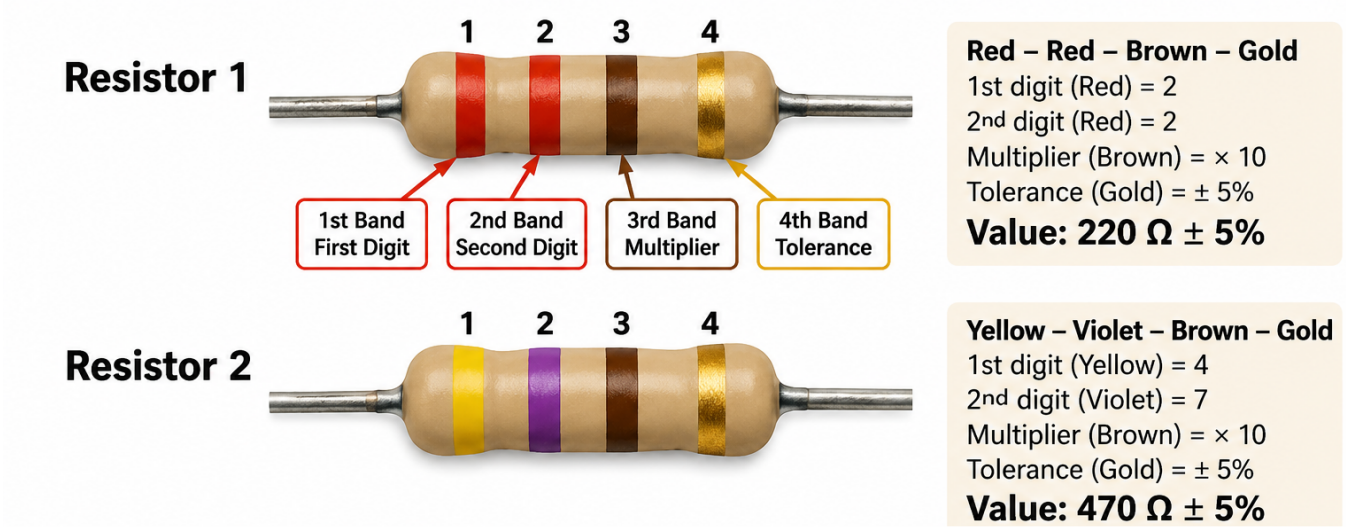
Mission: Find real resistors, decode their bands, check with a multimeter, and discover how resistor combinations change an LED circuit.

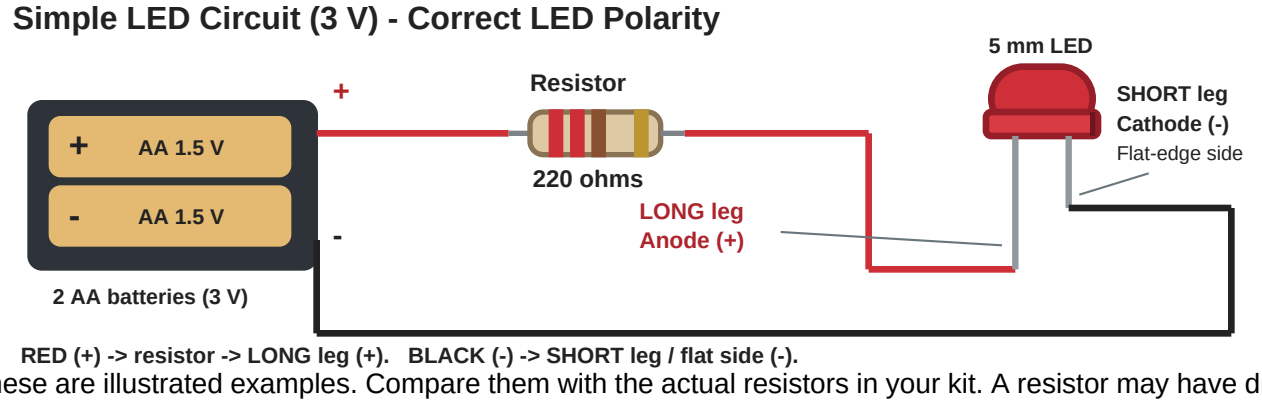
Your equipment: 2 AA batteries in a holder (~3 V), red/yellow 5 mm LED, resistors, multimeter, and connecting leads.

Safety: Teacher checks wiring before battery connection. Never connect the battery ends directly together. Disconnect the battery before measuring resistance. Always keep a current-limiting resistor in series with an LED.

STEP 1 — Find the resistor

Hold the gold tolerance band on the RIGHT. Read the other three bands from left to right. Use your own six-band chart as a reference.





These are illustrated examples. Compare them with the actual resistors in your kit. A resistor may have different body colors or band spacing.

STEP 2 — Identify and measure

Locate each resistor by color. Put the multimeter on Ω mode. Touch one probe to each resistor lead while the resistor is disconnected from the battery.

Find this value	4-band colors (left → right)	Measured Ω
220 Ω	Red / Red / Brown / Gold	_____
330 Ω	Orange / Orange / Brown / Gold	_____
470 Ω	Yellow / Violet / Brown / Gold	_____
1 kΩ	Brown / Black / Red / Gold	_____

Draw one real resistor you found and label its four bands:

_____

_____

STEP 3 — Build one LED circuit

Follow the circuit illustration on page 1: battery + → 220 Ω resistor → LED anode (+) → LED cathode (–) → battery –. The long LED leg is usually the anode; the flat side of the case indicates the cathode.

Have your teacher check it, then connect the batteries. Repeat using a 330 Ω and then 470 Ω resistor, one at a time.

Resistor	LED brightness (bright / medium / dim / off)	What did you notice?
220 Ω	_____	_____
330 Ω	_____	_____
470 Ω	_____	_____

STEP 4 — Two resistors: series versus parallel

Before connecting the LED: Disconnect the battery. Connect two 470 Ω resistors end-to-end (series). Measure resistance across the two free ends. Next connect both left ends together and both right ends together (parallel). Measure resistance across the joined ends.

Arrangement	Predict total Ω	Measure total Ω
One 470 Ω resistor	_____	_____
Two 470 Ω in series	_____	_____

Arrangement	Predict total Ω	Measure total Ω
Two 470 Ω in parallel	_____	_____

To observe brightness, reconnect the LED circuit with a **fixed 220 Ω protective resistor in series with the LED**, then insert each test network in series with that protected LED circuit. Ask your teacher to check before powering.

What changed? _____

_____

STEP 5 — Explain

1. Calculate theoretical equivalent resistance for two 470 Ω resistors in series and parallel.

_____

_____

2. Using 3.0 V and an assumed red LED drop of 2.0 V, estimate current with 220 Ω and 470 Ω series resistors.

_____

_____

3. Explain why LED brightness is an imperfect current measurement, and identify two experimental uncertainties.

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